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Module 5

Quality & Maintenance

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Module-5

- Software Maintenance
- Critical Systems Validation
- Quality Management
- Process Improvement
- Risk Management
- Configuration Management

Software Maintenance



- Software maintenance is the process of changing, modifying, and updating software to ensure that the software continues to meet the needs of the users over time.
- Software maintenance is a critical part of the software development life cycle (SDLC) and it is a continuous process that occurs throughout the entire life cycle of the software.
- Software maintenance is a process of modifying a software system after it has been delivered to the customer.
- Software maintenance is done after the product has launched for several reasons including improving the software overall, correcting issues or bugs, to boost performance, and more.
- The goal of software maintenance is to keep the software system working correctly, efficiently, securely for improving the system's functionality, performance, and reliability.

Software Maintenance



- It is also important to consider the cost and effort required for software maintenance when planning and developing a software system.
- It is important to have a well-defined maintenance process, which includes testing, validation, version control, and communication with stakeholders.
- Software maintenance can involve modifying the software code, as well as its documentation, user manuals, and training materials.
- Software developers constantly need to be on the lookout to both correct and improve their software to remain competitive and relevant.
- Software maintenance can also involve upgrading the software to a new version or platform. This can be necessary to keep up with changes in technology and to ensure that the software remains compatible with other systems.

Need of Software Maintenance



Software Maintenance must be performed in order to:

- Correct faults.
- Improve the design.
- Implement enhancements.
- Interface with other systems.
- Accommodate programs so that different hardware, software, system features, and telecommunications facilities can be used.
- Requirement of user changes.
- Improve performance

Key Aspects of Software Maintenance

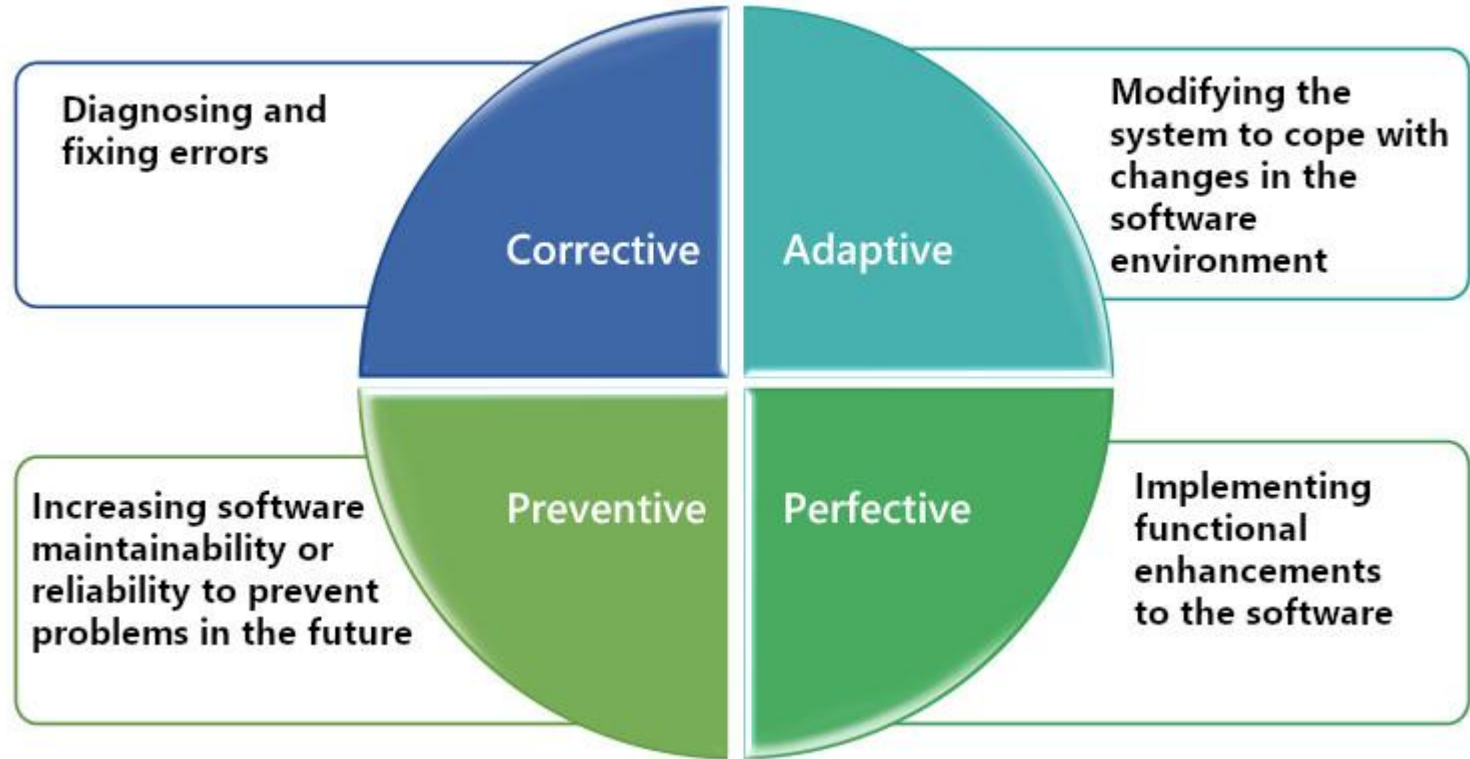
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- **Bug Fixing:** The process of finding and fixing errors and problems in the software.
 - **Enhancements:** The process of adding new features or improving existing features to meet the evolving needs of the users.
 - **Performance Optimization:** The process of improving the speed, efficiency, and reliability of the software.
 - **Porting and Migration:** The process of adapting the software to run on new hardware or software platforms.
 - **Re-Engineering:** The process of improving the design and architecture of the software to make it more maintainable and scalable.
 - **Documentation:** The process of creating, updating, and maintaining the documentation for the software, including user manuals, technical specifications, and design documents.

Types of Software Maintenance

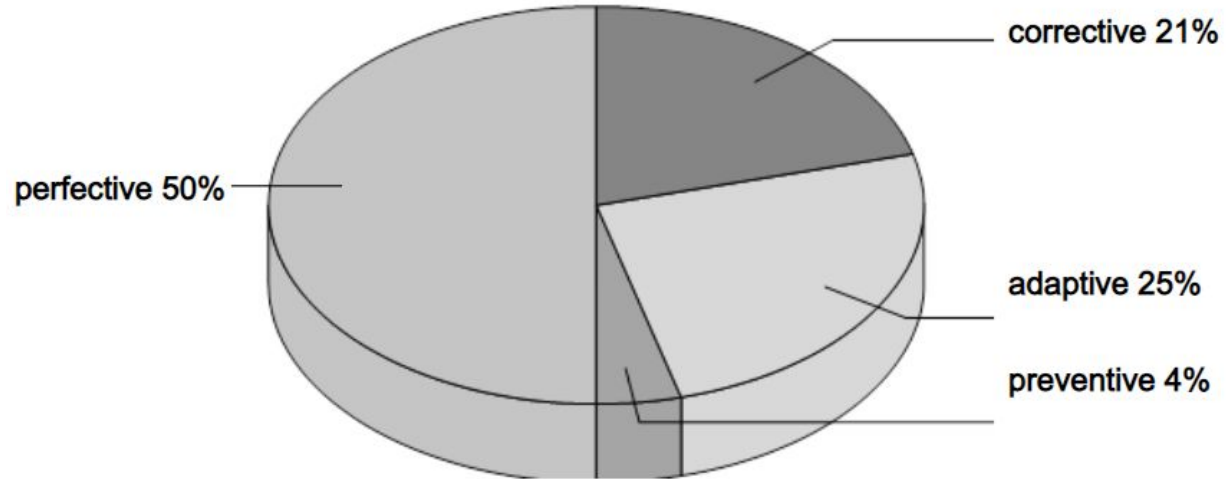


- **Corrective Maintenance:** This involves fixing errors and bugs in the software system observed while the system is in use, or to enhance the performance of the system.
- **Adaptive Maintenance:** This involves modifying and updating the software system to adapt it to changes in the environment, such as changes in hardware or software, government policies, and business rules.
- **Perfective Maintenance:** This involves improving functionality, performance, and reliability, and restructuring the software system to improve changeability.
- **Preventive Maintenance:** This involves taking measures to prevent future problems, such as optimization, updating documentation, reviewing and testing the system, and implementing preventive measures such as backups.

Types of Software Maintenance



Types of Software Maintenance



Software Maintenance Process

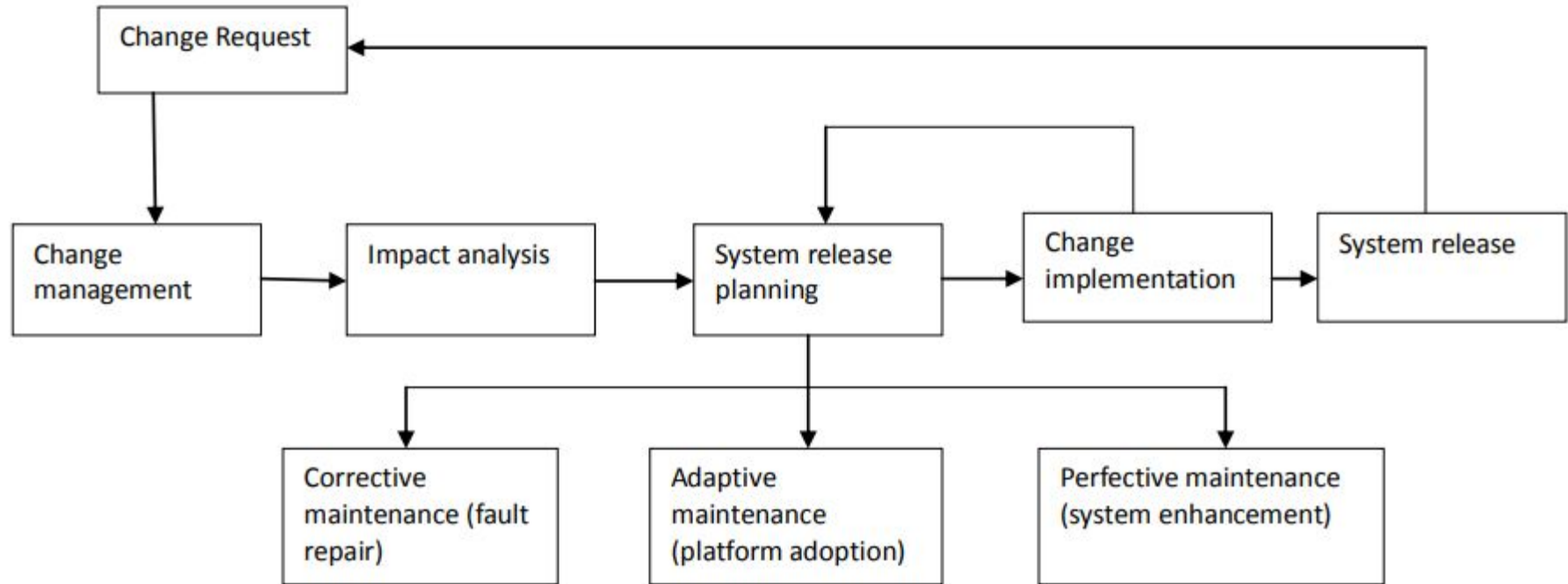


Figure 5.1: Maintenance Process

Issues in Software Maintenance

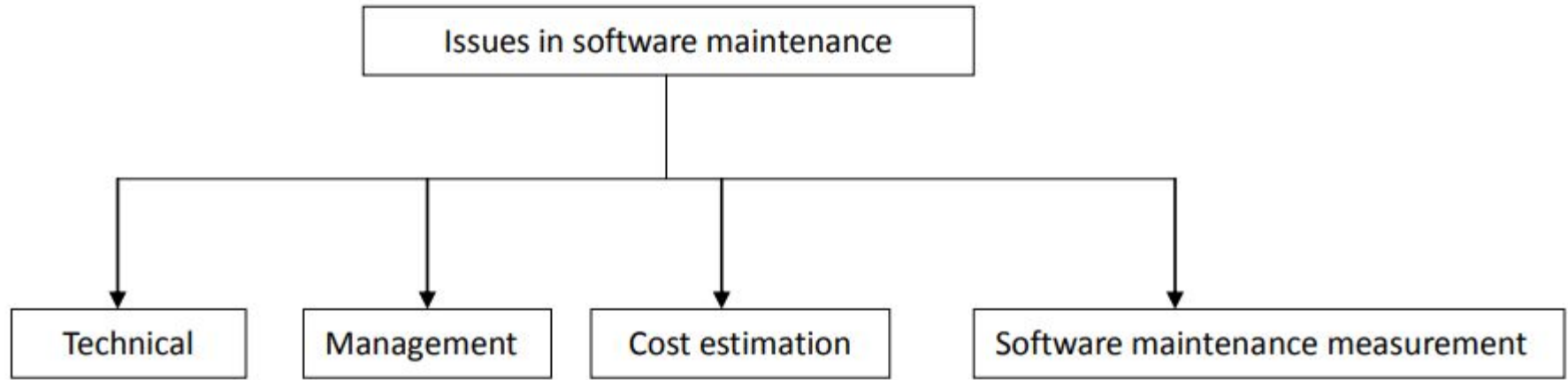


Figure 5.2: Software Maintenance Issues

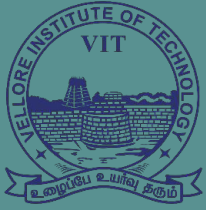


Advantages of Software Maintenance

- Improved Software Quality
- Enhanced Security
- Increased User Satisfaction
- Extended Software Life
- Cost Savings
- Better Alignment with business goals
- Competitive Advantage
- Improved Collaboration
- Improved Scalability

Disadvantages of Software Maintenance

- Cost
- Schedule disruptions
- Complexity
- Risk of introducing new bugs
- User resistance
- Compatibility issues
- Lack of documentation
- Technical debt
- Skill gaps



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Critical Systems Validation



Critical Systems Validation

Validating the reliability, safety and security of computer-based systems



Validation Perspectives

- **Reliability validation**
 - Does the measured reliability of the system meet its specification?
 - Is the reliability of the system good enough to satisfy users?
- **Safety validation**
 - Does the system always operate in such a way that accidents do not occur or that accident consequences are minimised?
- **Security validation**
 - Is the system and its data secure against external attack?



Reliability validation

- Reliability validation involves exercising the program to assess whether or not it has reached the required level of reliability
- Cannot be included as part of a normal defect testing process because data for defect testing is (usually) atypical of actual usage data
- Statistical testing must be used where a statistically significant data sample based on simulated usage is used to assess the reliability



Safety validation

- Design validation
 - Checking the design to ensure that hazards do not arise or that they can be handled without causing an accident.
- Code validation
 - Testing the system to check the conformance of the code to its specification and to check that the code is a true implementation of the design.
- Run-time validation
 - Designing safety checks while the system is in operation to ensure that it does not reach an unsafe state.



Security validation

- Security validation has something in common with safety validation
- It is intended to demonstrate that the system cannot enter some state (an unsafe or an insecure state) rather than to demonstrate that the system can do something
- However, there are differences
 - Safety problems are accidental; security problems are deliberate
 - Security problems are more generic; Safety problems are related to the application domain



Validation techniques

- Static techniques
 - Design reviews and program inspections
 - Mathematical arguments and proof
- Dynamic techniques
 - Statistical testing
 - Scenario-based testing
 - Run-time checking
- Process validation
 - Design development processes that minimise the chances of process errors that might compromise the dependability of the system



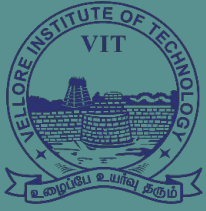
Static validation techniques

- Static validation is concerned with analyses of the system documentation (requirements, design, code, test data).
- It is concerned with finding errors in the system and identifying potential problems that may arise during system execution.
- Documents may be prepared (structured arguments, mathematical proofs, etc.) to support the static validation



Dynamic validation techniques

- These are techniques that are concerned with validating the system in execution
 - Testing techniques - analysing the system outside of its operational environment
 - Run-time checking - checking during execution that the system is operating within a dependability ‘envelope’



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Quality Management





Software Quality Management

- Concerned with ensuring the required level of quality is achieved in a software product
- Involves the definition of appropriate quality standards and the definition of procedures to ensure that these standards are followed
- Works best when a 'quality culture' is created where quality is seen as everyone's responsibility



Quality Definition

- Quality means that a product satisfies the demands of its specifications
- It also means achieving a high level of customer satisfaction with the product
- In software systems this is difficult
 - customer quality requirements (e.g. efficiency or reliability) often conflict with developer quality requirements (e.g. maintainability or reusability)
 - software specifications are often incomplete, inconsistent, or ambiguous



Quality Management Activities

- Quality assurance
 - establishing organizational quality standards and procedures
- Quality planning
 - selecting and modifying applicable quality standards and procedures for a particular project
- Quality control
 - ensuring quality standards and procedures are followed by development team

Note: Quality management should be separated from project management to ensure independence.



Quality Standards

- Key to effective quality management
- Product standards define the characteristics exhibited by all components (e.g. programming style issues)
- Process standards describe how a software process is to be implemented
- Should encapsulate best practices - this helps avoid repeating past mistakes
- Provide continuity by giving new team members a means to understand the organizational priorities



Process and Product Standards

Product Standards

Design review form

Document naming
standards

Function prototype format

Programming style
standards

Project plan format

Change request form

Process Standards

Design review guidelines

Document submission
procedures

Version release process

Project plan approval
procedure

Change control process

Test data recording
procedures



Quality Standards Development

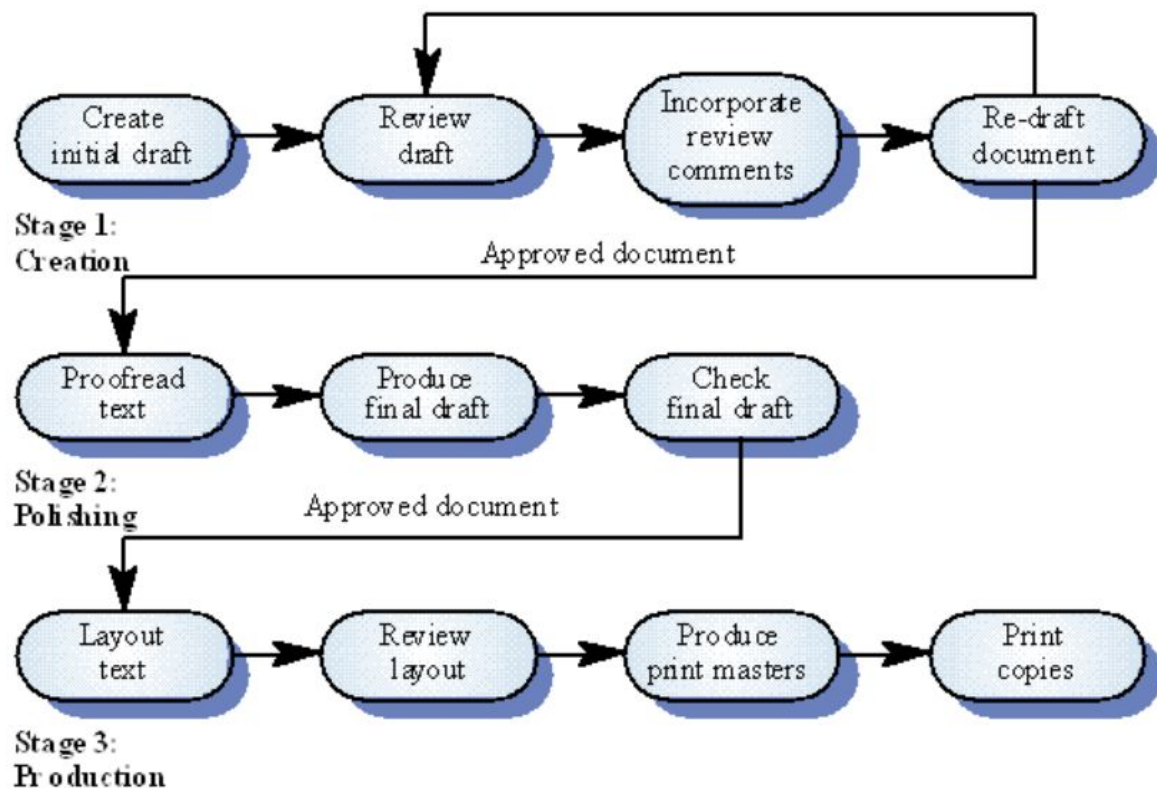
- Should involve practitioners in their development
- Engineers must understand the rationale behind each standard
- Standards must be reviewed and revised regularly to avoid obsolescence and credibility problems with practitioners
- Detailed standards need tool support to eliminate the “too much clerical work” excuse for not following the standards



Documentation Standards

- Documentation process standards
 - describe how documents are to be developed, validated, and maintained
- Document standards
 - concerned with document content, structure, and appearance
- Document interchange standards
 - specify how documents are to be stored and shared between different documentation systems

Documentation process

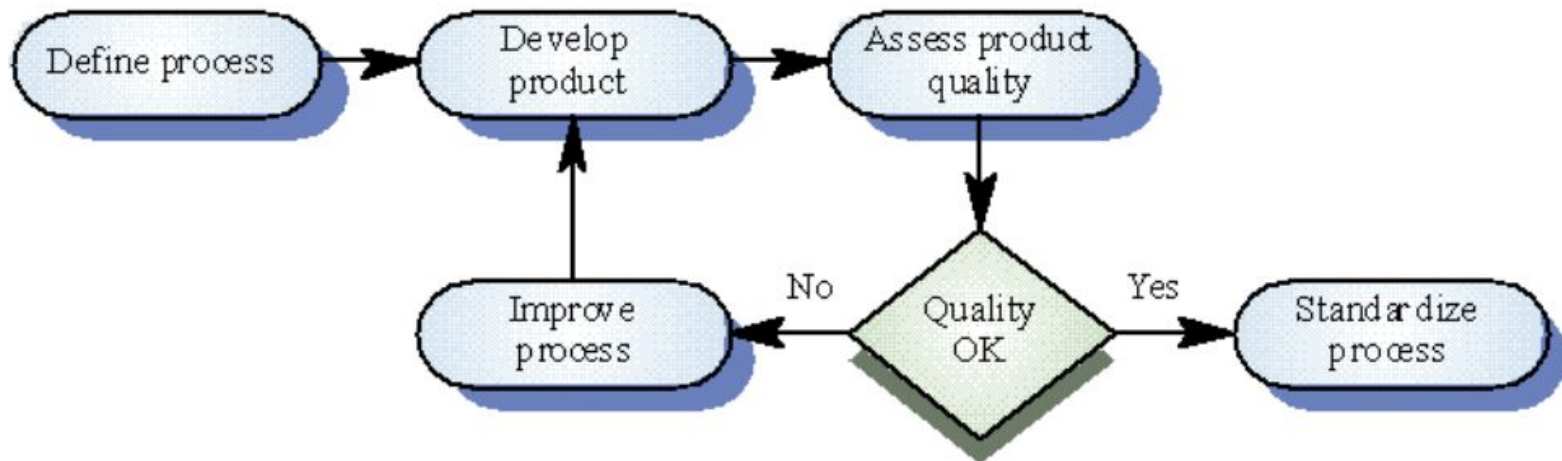




Process-Based Quality

- Product quality is influenced by the quality of its production process
- This relationship is easy to see in the manufacture of goods, it is more complex for software production because
 - the application of individual skills and experience is particularly important in software development
 - external factors (e.g. application novelty or need to accelerate schedule) are more likely to impair quality

Process-Based Quality Activities





Quality Plan

- Identifies the most significant quality attributes appropriate for the product
- Defines the assessment process in detail for each quality attribute
- Indicates which organization standards should be applied and defines new standards as necessary



Quality Plan Components

- Product introduction
- Product plans
- Process descriptions
- Quality goals
- Risks and risk management



Software Quality Attributes

- Safety
- Security
- Reliability
- Resilience
- Robustness
- Understandability
- Testability
- Adaptability
- Modularity
- Complexity
- Portability
- Usability
- Accessibility
- Reusability
- Efficiency
- Learnability



Quality Control

- Examines the software development process to ensure that all relevant procedures and standards are being followed
- Two basic approaches
 - quality reviews
 - software measurement and assessment



Reviews

- Reviews are the principle means of validating both process and product quality
- Basic procedure is to have a group of people examine a process artifact to find potential problems
- Common software review types include
 - defect inspection and removal (product)
 - progress reviews (product and process)
 - quality reviews (product and standards)



Quality Review Process

- Select a review team
- Arrange a time and place for the review
- Distribute documents to review
- Conduct the review
- Complete the review forms
- Decide whether to approve artifacts or have them reworked and review them again



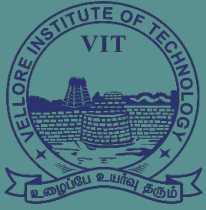
Software Measurement and Metrics

- Software measurement is concerned with deriving a numeric value for an attribute of a software product or process
- This allows for object comparisons between techniques or processes
- The systematic use of measurement is essential to process improvement programs



Software Metric

- Any type of measurement that relates to a software system, process, or document
 - LOC, person-months, function points
- Metrics allow for the quantification of the software or a software process
- May be used to predict product attributes or to control the software process



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Software Configuration Management

Software Configuration Management



- Software configuration management is a set of activities carried out for identifying, organizing and controlling changes throughout the life cycle of computer software.
- It is a process to systematically manage, organize, and control the changes in the documents, codes, and other entities.
- The primary goal is that, “**Changes must be managed and controlled**” in order to improve quality and reduce errors.
- The SCM system controls the basic components such as software objects, program code, test data, test output, design documents and user manuals.
- Hence software configuration management is a quality assurance activity that is applied throughout the software process.
- SCM helps to eliminate the confusion often caused by miscommunication among team members.

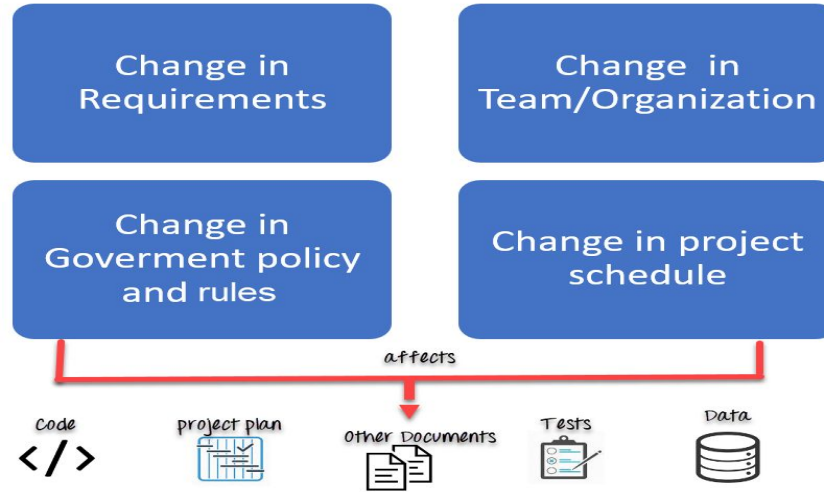
Why Software Configuration Management?



The primary reasons for Implementing Technical Software Configuration Management System are:

- There are multiple people working on software which is continually updating.
- It may be a case where multiple version, branches, authors are involved in a software config project, and the team is geographically distributed and works concurrently
- Changes in user requirement, policy, budget, schedule need to be accommodated.
- Software should able to run on various machines and Operating Systems
- Helps to develop coordination among stakeholders
- SCM process is also beneficial to control the costs involved in making changes to a system

Why Software Configuration Management?



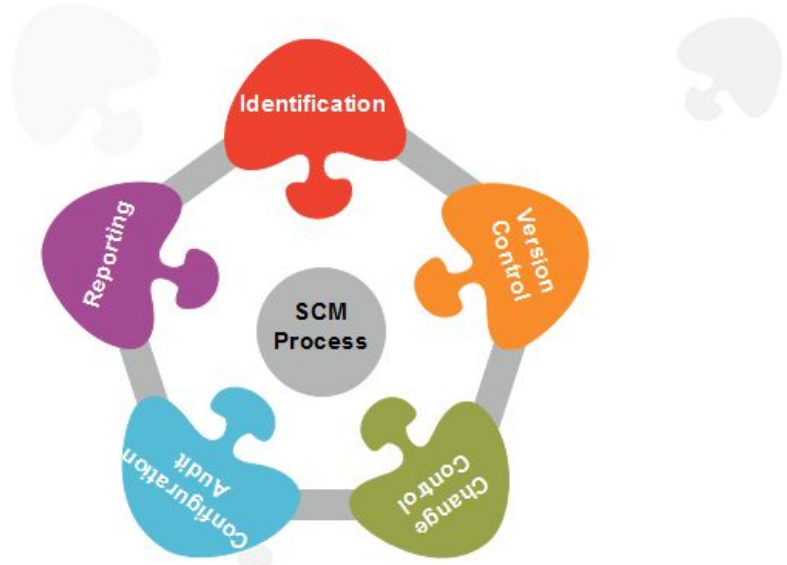
Any change in the software configuration Items will affect the final product. Therefore, changes to configuration items need to be controlled and managed.

Software Configuration Management Process

The SCM process defines a number of tasks:

- Software Configuration Identification
- Software Configuration Control
 - Version Control
 - Change Control
- Software Configuration Auditing
- Software Configuration Status Reporting

Software Configuration Management Process



Software Configuration Management Process

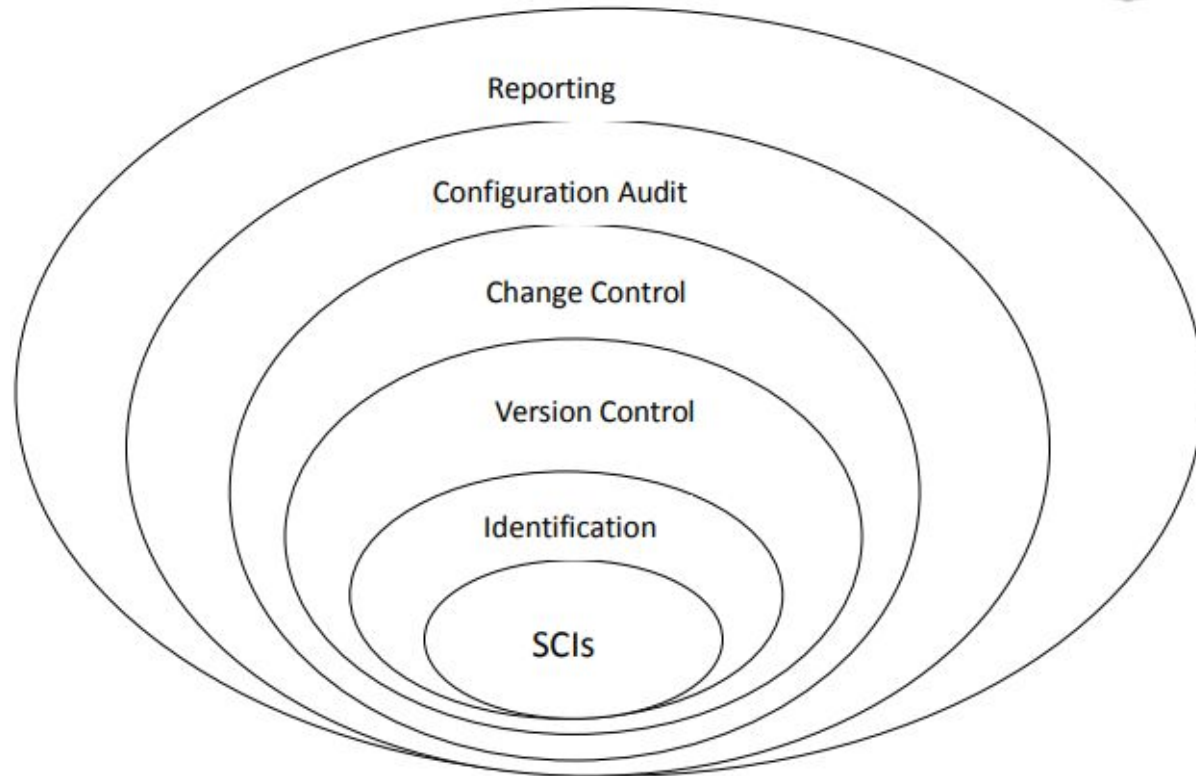


Figure 5.3: SCM Process

Software Configuration Management Process

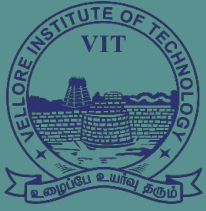
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1. **Configuration Identification:** Configuration identification is a method of determining the scope of the software system.
 2. **Baseline:** A baseline is a formally accepted version of a software configuration item. It is designated and fixed at a specific time while conducting the SCM process.
 3. **Change Control:** Change control is a procedural method which ensures quality and consistency when changes are made in the configuration object. In this step, the change request is submitted to the software configuration manager.
 4. **Configuration Status Accounting:** Configuration status accounting tracks each release during the SCM process. This stage involves tracking what each version has and the changes that lead to this version.
 5. **Configuration Audits and Reviews:** Software Configuration audits verify that all the software product satisfies the baseline needs. It ensures that what is built is what is delivered.

Advantages of Software Configuration Management

- 
1. Improved productivity and efficiency by reducing the time and effort required to manage software changes.
 2. Reduced risk of errors and defects by ensuring that all changes are properly tested and validated.
 3. Increased collaboration and communication among team members by providing a central repository for software artifacts.
 4. Improved quality and stability of software systems by ensuring that all changes are properly controlled and managed.

Disadvantages of Software Configuration Management

- 
1. Increased complexity and overhead, particularly in large software systems.
 2. Difficulty in managing dependencies and ensuring that all changes are properly integrated.
 3. Potential for conflicts and delays, particularly in large development teams with multiple contributors.



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Process Improvement



Software process improvement



- Software process improvement is the art and science of changing an organization's software process in order to build better software.
- Software process improvement always involves looking at the big picture.
- This means writing down the entire software process as a whole and making sure that it is followed on each project.
- It involves looking at entire projects and identifying areas that can be improved.

Software process improvement



Frameworks and Methodologies

- Models and certifications that help assess the state of the organization's process and serve as a framework for improving that process.
 - CMM, ISO 9000 and Six Sigma are frameworks
- There are also methodologies that an organization can adopt that describe the complete set of activities, roles, and work products needed to build software.
 - Rational Unified Process and Extreme Programming are methodologies.

Capability Maturity Model (CMM)



- The Software Process Capability Maturity Model (CMM) deals with the capability of software organizations to consistently and predictably produce high quality products.
- It is closely related to software process, quality management, and process improvements.
- The CMM defines the characteristics of a mature, capable process in a way that can be measured and compared to processes at other organizations.
- Process capability is the inherent ability of a process to produce planned results.
- A software engineering process group within the organization identifies problems, inefficiencies and defines practices to address them.

Process Maturity Model

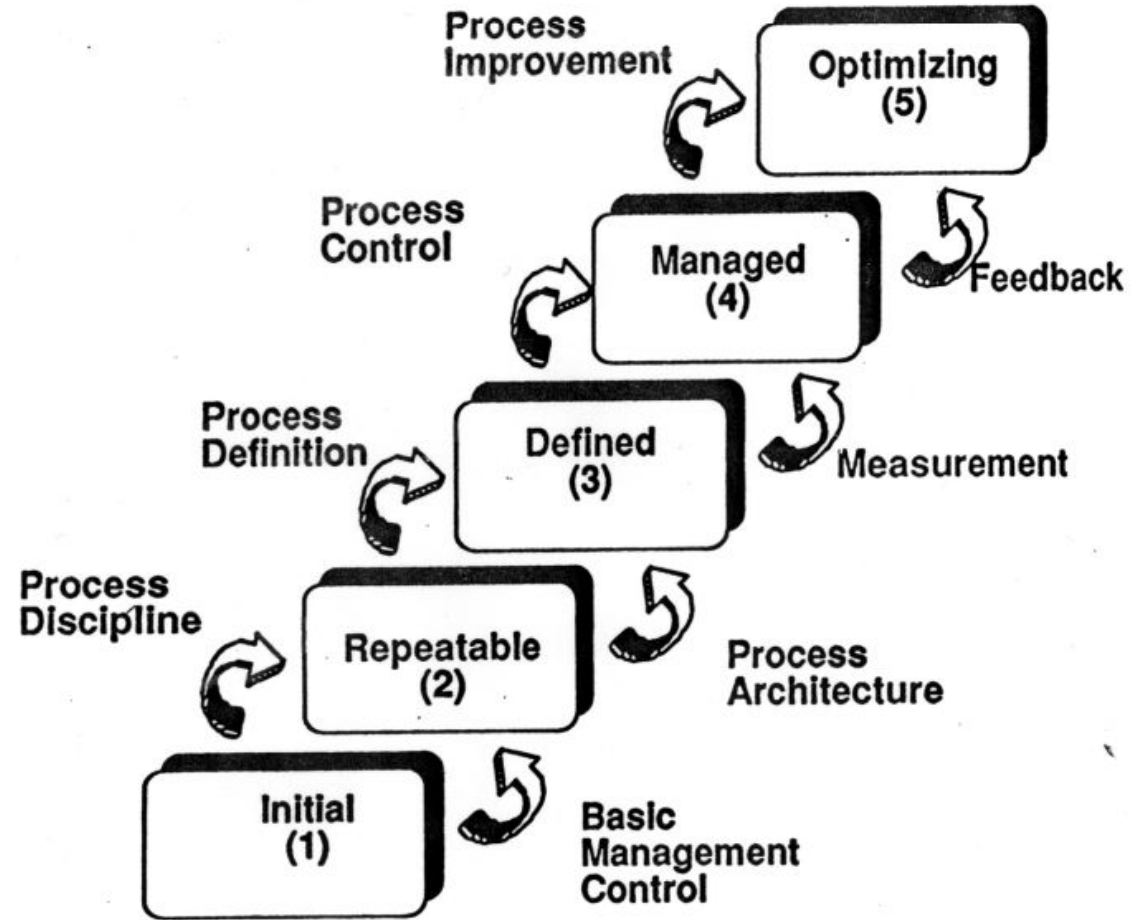


Figure 4-1: The Five Levels of Software Process Maturity

Process Maturity Model

- At the initial level (level 1), an organization can be characterized as having an ad hoc process.
- Typically, the organization operates without formalized procedures, cost estimates, and project plans.
- Tools are not well integrated with the process, nor are they uniformly applied.

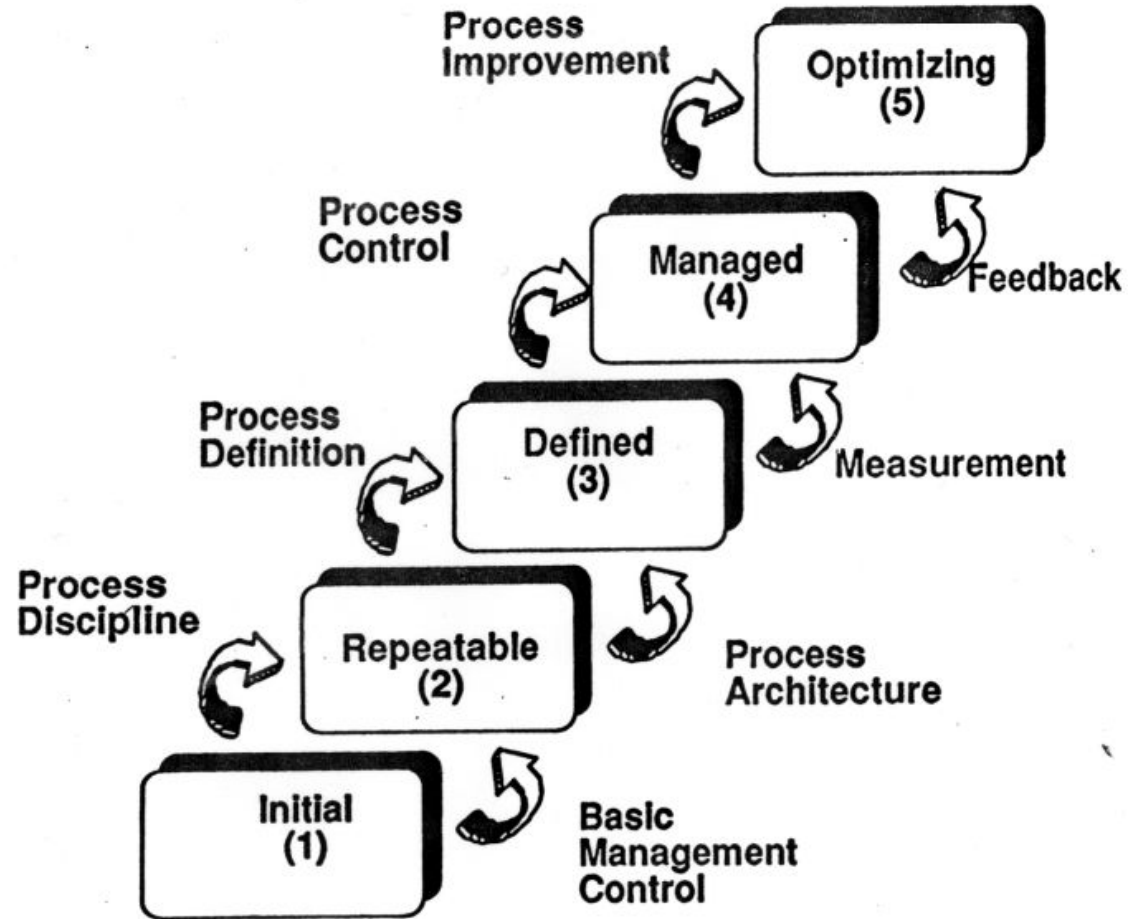


Figure 4-1: The Five Levels of Software Process Maturity

Process Maturity Model

- An organization at the repeatable level (level 2) has established basic project controls.
- It comprises of project management, product assurance, and change control.
- The strength of the organization stems from its experience, but it faces major risks with new challenges.

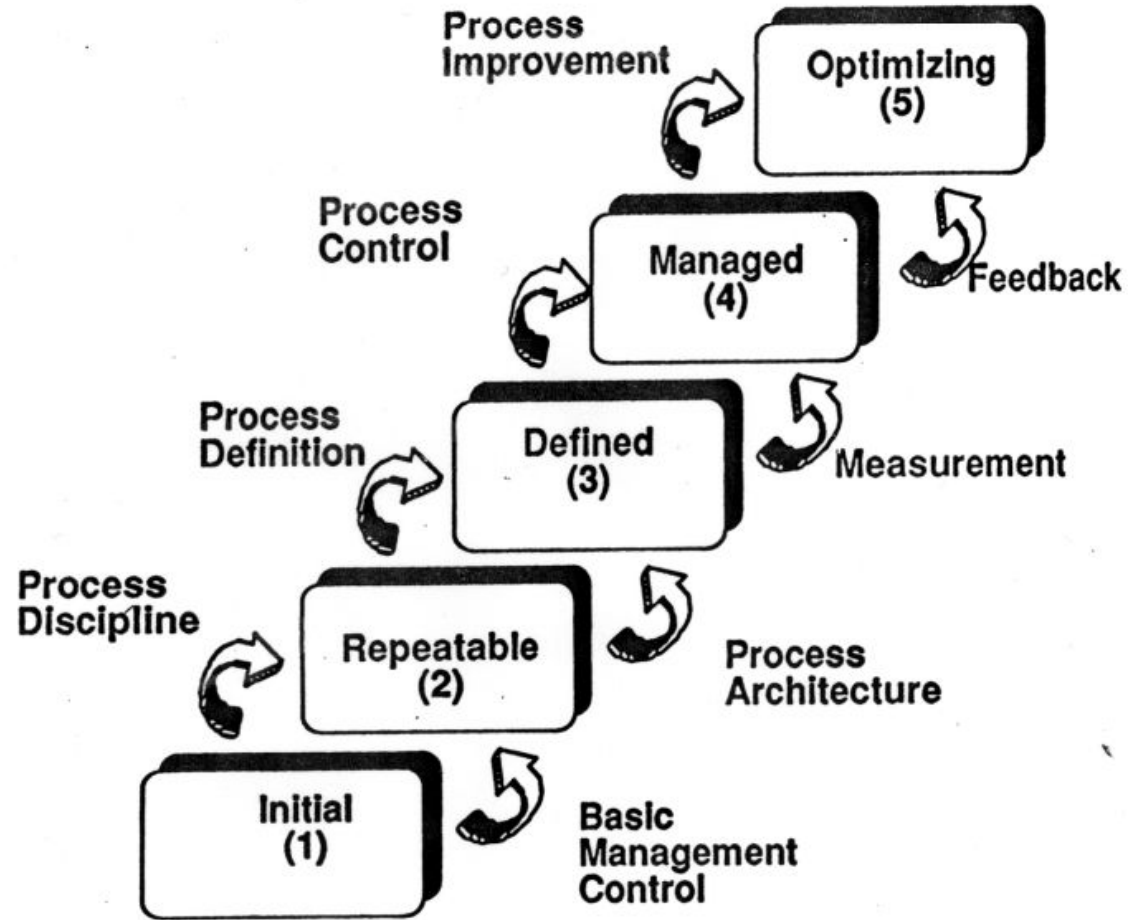


Figure 4-1: The Five Levels of Software Process Maturity

Process Maturity Model

- At the defined level (level 3), the organizations examine the process and decide how to improve it.
- A Software Engineering Process Group (SEPG) has been established to focus and lead the process improvement efforts.
- It facilitates software engineering methods and technologies.

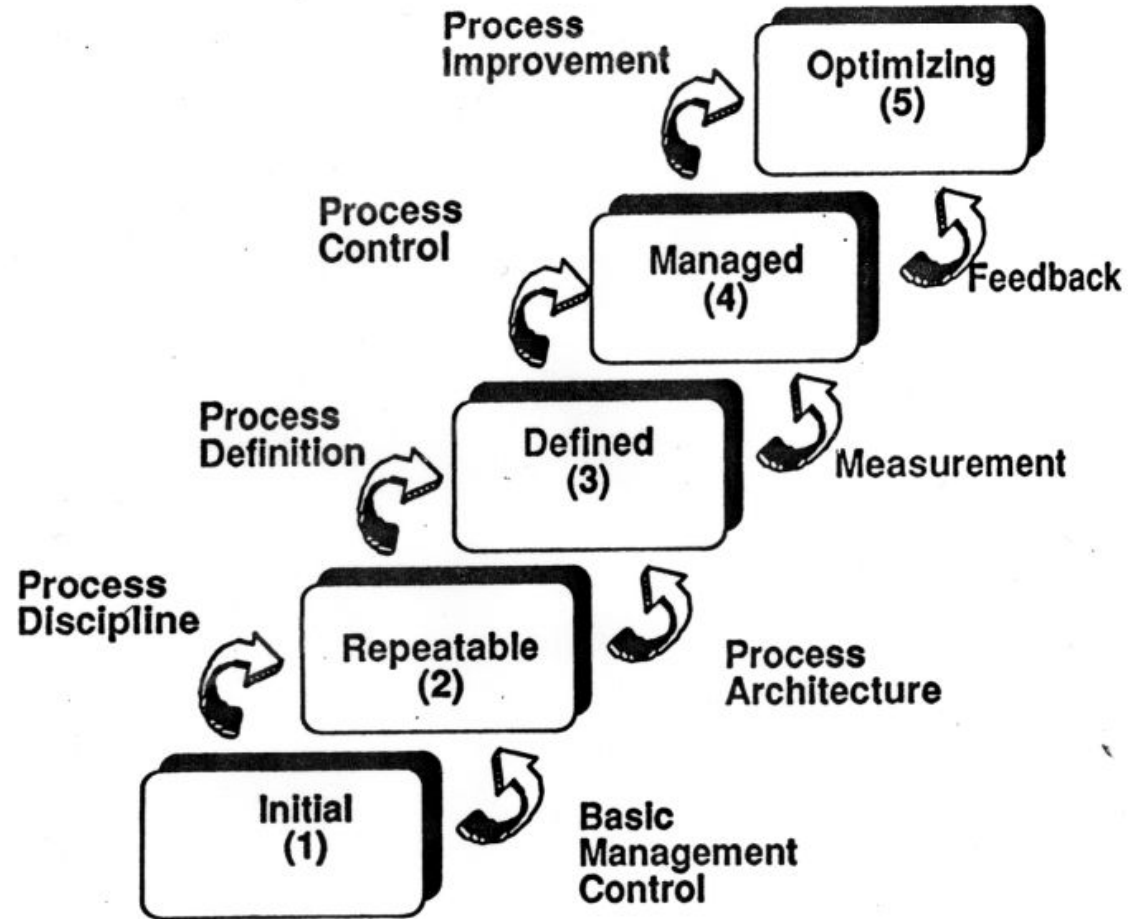


Figure 4-1: The Five Levels of Software Process Maturity

Process Maturity Model

- When the process is defined, it can be examined and improved at managed level (level 4).
- Organizations establish a set of quality and productivity measurements at this level.
- A feedback cycle is used to continually plan, implement, and track process improvements.

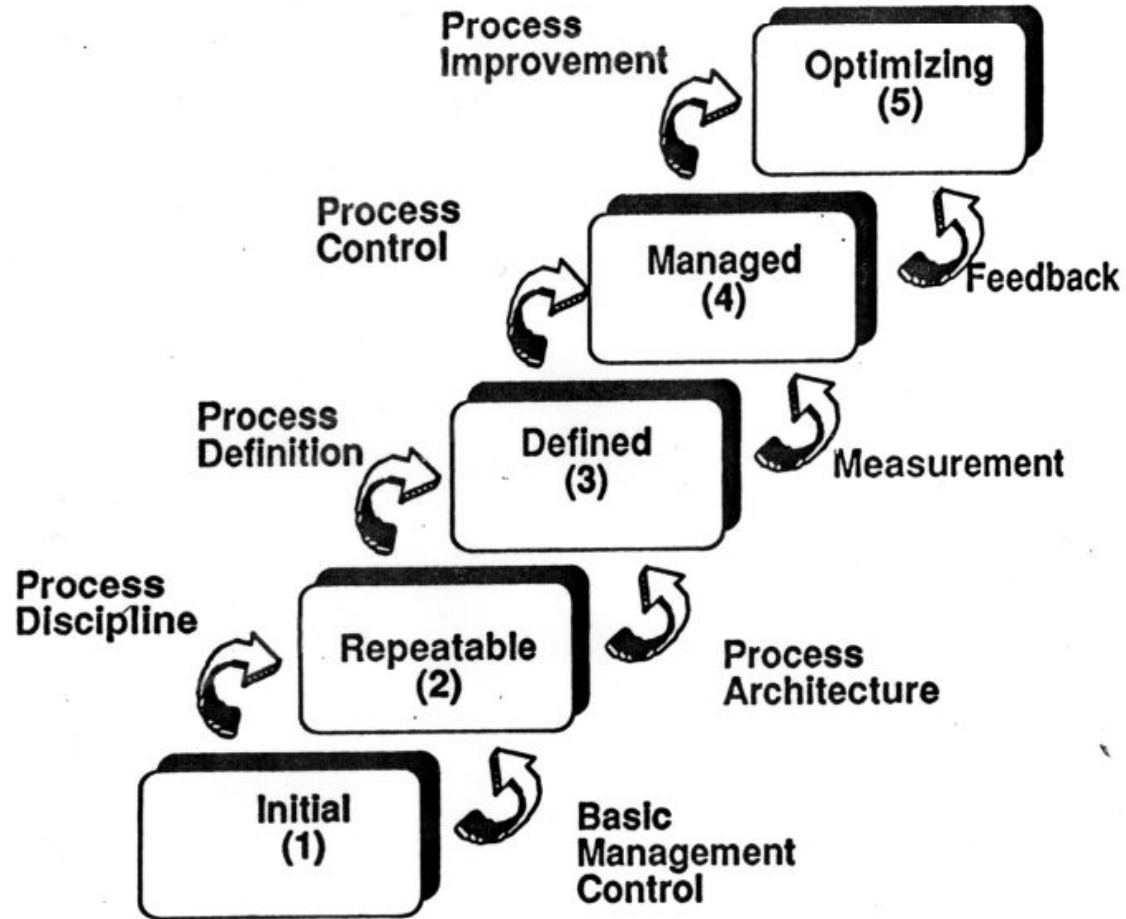


Figure 4-1: The Five Levels of Software Process Maturity

Process Maturity Model

- At the optimizing level (level 5) the organization has the means to identify its weakest process elements and strengthen them.
- Data are available to justify applying technology to critical tasks.
- Evidence is available on process effectiveness.
- Additional activity at the optimizing level is defect cause analysis and defect prevention.

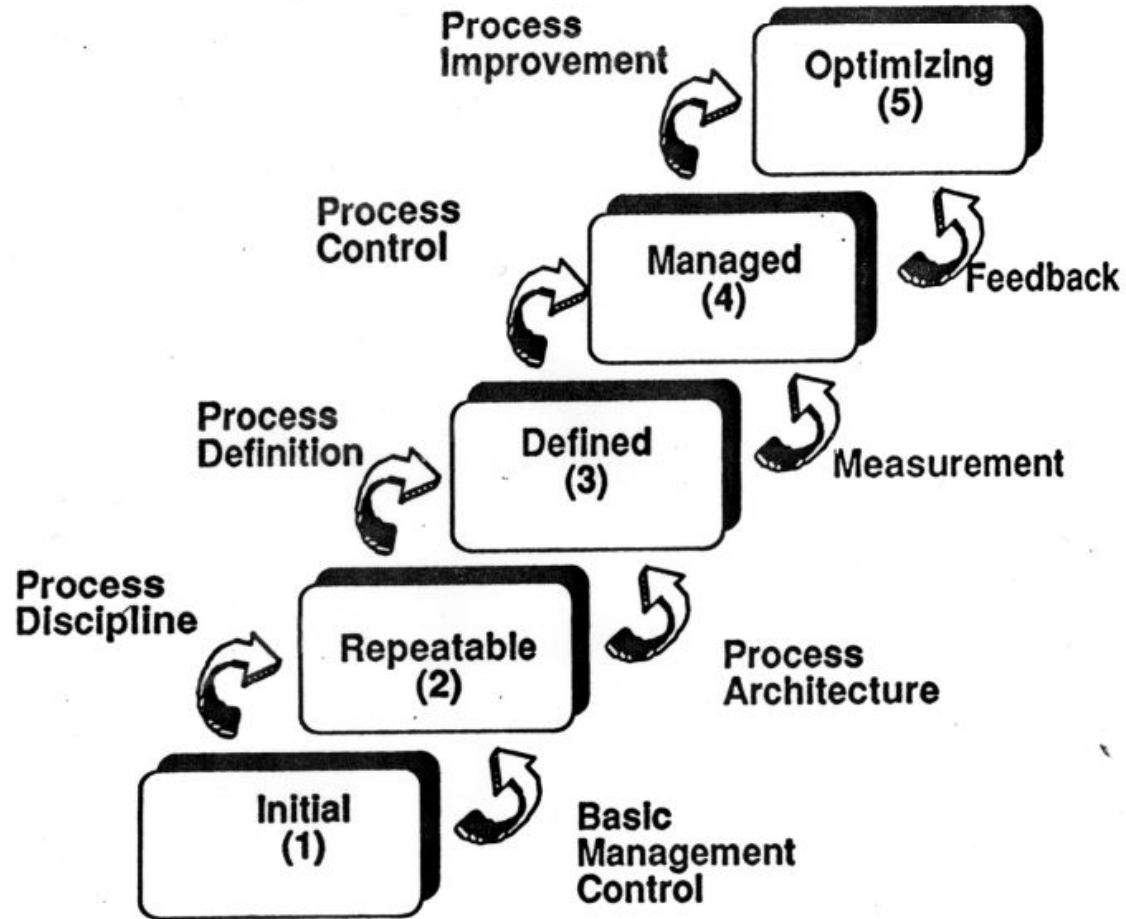


Figure 4-1: The Five Levels of Software Process Maturity

ISO 9000



ISO 9000 is a family of *quality management* standards defined by the International Standards Organization.

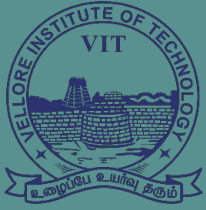
It is based on core principles:

- Organizations must focus on their customers by understanding current and future customer needs.
- Leaders within the organization must create and maintain an environment in which people can become involved and fulfill the organization's objectives.
- People at all levels are important to the organization.
- Activities and resources are best managed as a process.
- Organizations have many interrelated processes, which must be understood and managed as a system.
- The organization should continually improve its performance.
- Decisions should be well informed and based on real data and information.
- An organization and its suppliers are in a mutually beneficial relationship.

Six Sigma



- Six Sigma is an approach to improving quality in manufacturing and business processes.
- The Greek letter sigma refers to standard deviation—Six Sigma means “six standard deviations from the mean.”
- DMAIC is a five-phase approach to Six Sigma improvement
 - Define opportunities,
 - Measure performance,
 - Analyze opportunity,
 - Improve performance,
 - Control performance



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Risk Management



Risk Management

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- A risk is a probable problem- **it might happen or it might not.**
 - Risk management involves all activities related to identifying, analysing and making provision for predictable and non-predictable risks in the project
 - Risk management is a sequence of steps that help a software team to understand, analyze, and manage uncertainty.

Risk may include the following:

- **Experienced staff leaving the project and new staff coming in.**
- **Change in organizational management.**
- **Requirement change or misinterpreting requirement.**
- **Under-estimation of required time and resources.**
- **Technological changes, environmental changes, business competition**

Risk Management Process



Risk management performed in following stages:

- **Risk identification:** In this phase all possible risks are anticipated and a list of potential risks are prepared.
- **Risk analysis:** After risk identification, a list is prepared in which risks are prioritized.
- **Risk planning:** The risk avoidance or risk minimization plan is prepared in this phase.
- **Risk monitoring:** Identified risks must be mitigated. Hence risk mitigation plan must be prepared once the risks are discovered.

Risk Management Process

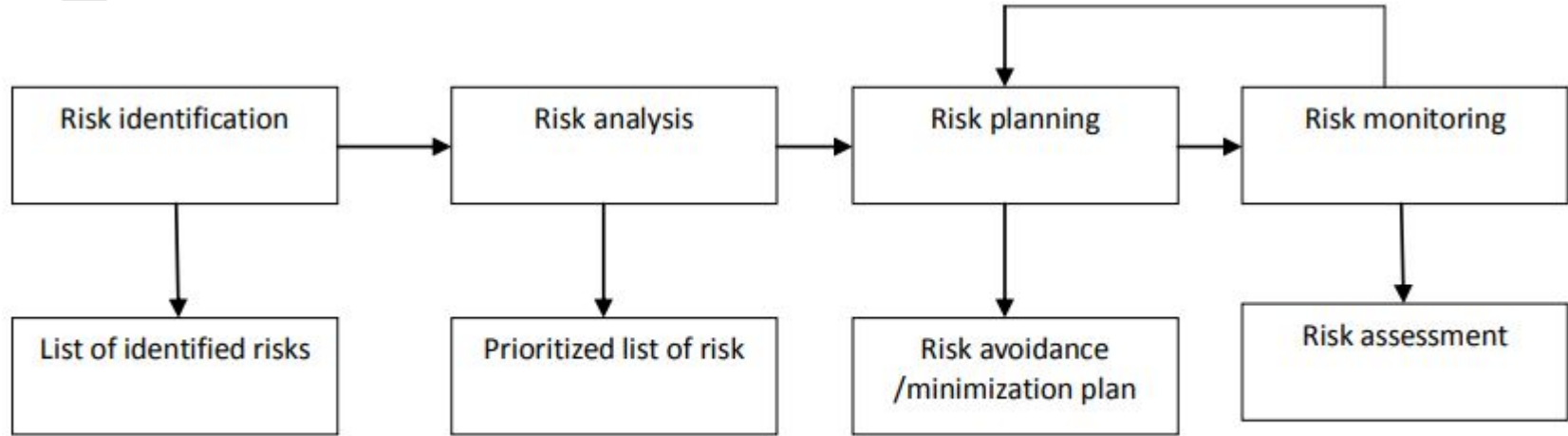


Figure 5.9: Risk management process

Risk Management Process



There are mainly 3 classes of risks that may affect a computer code project:

Project Risks: Project risks concern funds, schedules, personnel, resources, and customer-related issues.

Technical Risks: Technical risks concern implementation, interfacing, testing, and maintenance issues.

Business Risks: This type of risk embodies the risks of building a superb product that nobody needs, losing funds or personal commitments, etc.

Other risk categories

- **Known risks:** Those risks that can be uncovered after careful assessment.
- **Predictable risks:** Risks that are hypothesized from previous project experience.
- **Unpredictable risks:** Risks that can and do occur, but are extremely tough to identify in advance.

Risk Assessment



- Risk assessment simply means to describe the overall process or method to identify risk and problem factors that might cause harm.
- It is actually a systematic examination of a task or project to simply identify significant risks, problems, hazards, and then to find out control measures that you will take to reduce risk.
- The best approach is to prepare a set of questions that can be answered by project managers in order to assess overall project risks. These questions are shown below:
 - Is there a clear understanding of the requirements?
 - Is project scope stable?
 - Are there team members with the required skills?
 - Are project requirements stable?
 - Does technology used for software is known to developers?
 - Is the size of the team sufficient to develop the required product?
 - Is that all customers know the importance of the product/requirements of the system to be built?

Risk Mitigation



- Risk mitigation simply means to reduce effects and impact of risks that are harmful to project and Business continuity.
- Risk mitigation means preventing risks to occur (risk avoidance).
- It includes introducing measures and step taken into a project plan to mitigate, reduce, eliminate, or control risk.
- Following are measures and steps to be taken for mitigating risks:
 - Communicate with concerned staff to find probable risks.
 - Identify and eliminate all those causes and issues that can create risk before the beginning of project work.
 - Develop policy in an organization that will help to continue the project even though some staff leaves the organization.
 - Everybody in the project team should be aware of and familiar with current development activity.
 - Maintain corresponding documents in a timely manner. This documentation should be strictly followed as per standards set by the organization.
 - Conduct timely reviews in order to speed up work.
 - For conducting every critical activity during software development, provide additional staff is required.

Risk Assessment Metrix



- A Risk Matrix is a tool used in risk management to visualize and evaluate potential risks in an organized and systematic manner.
- It provides a graphical representation of the likelihood and potential impact of identified risks, allowing organizations to prioritize and allocate resources effectively to mitigate or prevent potential losses.
- The matrix categorizes risks based on their severity and probability, enabling decision makers to understand the overall risk exposure and develop informed strategies for risk mitigation.
- By using a Risk Matrix, organizations can proactively manage and control risks, improve their overall risk management process, and make informed decisions that will protect their interests and promote growth.

RISK ANALYSIS

		CONSEQUENCE				
LIKELIHOOD		1. INSIGNIFICANT Dealt with by in house first aid	2. MINOR Treated by medical professionals, hospital out patients	3. MODERATE Significant non permanent injury overnight hospital stay	4. MAJOR Extensive permanent injury eg. Loss of fingers, extended hospital stay	5. CATASTROPHIC Death, permanent disabling injury eg. Loss of hand, quadriplegia
	A. Almost certain to occur in most circumstances	MEDIUM 8	HIGH 16	HIGH 18	CRITICAL 23	CRITICAL 25
	B. Likely to occur frequently	MEDIUM 7	MEDIUM 10	HIGH 17	HIGH 20	CRITICAL 24
	C. Possibly and likely to occur at sometime	LOW 3	MEDIUM 9	MEDIUM 12	HIGH 19	HIGH 22
	D. Unlikely to occur but could happen	LOW 2	LOW 5	MEDIUM 11	MEDIUM 14	HIGH 21
	E. May occur but only in rare circumstances	LOW 1	LOW 4	LOW 6	MEDIUM 13	MEDIUM 15

Benefits of Risk Assessment Metrix



- **Prioritization of Risks**
- **Improved Communication**
- **Better Decision Making**
- **Increased Awareness**
- **Consistency**
- **Better Resource Allocation**
- **Improved Compliance**